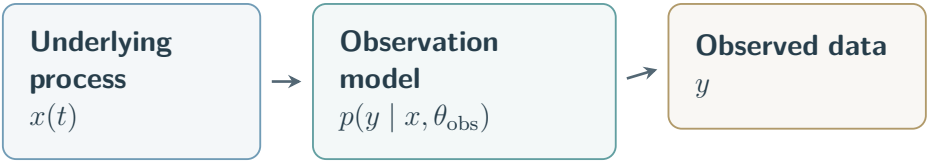
Models and measurements

An observation model describes how a scientific process is measured.



Health systems

Disease states
Detection / reporting delay
Cases / admissions / serology

Marine environments

Ecosystem state
Sampling / spatial aliasing
Surveys / eDNA / sensors

Human systems

Traits / population structure
Recovery / coding / sampling
Assemblages / language / genetics

Earth systems

Soil, fire, vegetation
Sensor placement / cloud
Indices / fluxes / transects

A shared mathematical form still requires domain-specific observation assumptions and independent validation. This is a conceptual map of research domains.